

1. Administrative & Study Identification

Full Study Title: An Accessorised Prefilled Syringe to an Autoinjector Pharmacokinetic Bridging Study of Tozorakimab
Short Title/Acronym: Tozorakimab APFS to AI Bridging Study
Protocol Number: CTIS2024-511840-22-00
NCT Number: NCT06908577
Sponsor Name: AstraZeneca
Investigational Product: Tozorakimab (MEDI3506) / Autoinjector (AI) and Accessorised Prefilled Syringe (APFS) Devices
Phase of Development: Phase I
Medical Monitor: AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To compare the pharmacokinetic (PK) exposures of a single subcutaneous (SC) dose of tozorakimab administered using an Autoinjector (AI) or an Accessorised Prefilled Syringe (APFS) in healthy participants.
Secondary Objectives: To evaluate the safety and tolerability of tozorakimab delivered via APFS versus the AI device.

2.2 Background & Rationale To establish a pharmacokinetic bridge between two different administration devices (APFS and AI) for the delivery of tozorakimab. Proving comparable exposure supports flexible dosing and device options for future clinical and commercial use in patients requiring tozorakimab.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Device comparison) **Blinding:** Open-label
Randomization: 1:1:1:1:1:1 to one of the 6 combinations of the devices (APFS or AI) and one of three injection sites (abdomen, thigh, or upper arm).
3.2 Planned Enrollment & Duration Target \$N\$: Not explicitly specified (Standard Phase 1 healthy volunteer cohort size) **Number of Sites:** Multiple centers (USA, Germany, UK) **Estimated Study Start:** 2024/2025 **Estimated Primary Completion:** 25-Nov-2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Healthy adults with suitable veins for cannulation or repeated venipuncture. 2. Body mass index (BMI)

between 19 and 30 kg/m2 inclusive, and weight of at least 55 kg and no more than 100 kg inclusive at screening and Day -1. 3. Intact normal skin without potentially obscuring tattoos, scars, etc., at the injection site. *(Females must have a negative pregnancy test and agree to use an approved highly effective contraception method until 16 weeks post-administration).*

4.2 Exclusion Criteria 1. History of any clinically significant disease or disorder which may either put the participant at risk or influence the results of the study. 2. Evidence of currently active tuberculosis (TB) disease or use of any TB drug treatment in the past 12 months, or latent TB infection. 3. Any clinically significant abnormalities on 12-lead electrocardiogram (ECG) at the screening visit and/or admission (Day -1).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single dose administration. **Route of Administration:** Subcutaneous (SC) injection. **Duration of Treatment:** Single SC dose on Day 1, with follow-up extending to Day 113 (16 weeks).

5.2 Concomitant & Prohibited Therapy Permitted: Generally restricted in Phase 1 trials; limited standard medications cleared by the Principal Investigator. **Prohibited:** TB drug treatments within the past 12 months; any medications likely to impact pharmacokinetic assessments.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to -2	Informed Consent, Medical History, ECG, Labs
Baseline	Day -1	Admission to Clinical Unit, Vital Signs, Contraception check
Treatment & Confinement	Day 1 to Day 9	Randomization, Single SC Dose via APFS or AI device, clinical unit monitoring
Outpatient Follow-up	Day 10 to Day 85	Sequential outpatient follow-up visits, PK and safety assessments
End of Study	Day 113 (Week 16)	Final follow-up PK and safety assessments, Exit Interview

(Note: The treatment period includes a stay in the clinical unit for up to 9 days, followed by an outpatient follow-up period until Day 85, and a final visit on Day 113).

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Pharmacokinetic (PK) exposures (e.g., Cmax, AUC) of a single subcutaneous (SC) dose of

tozorakimab. **Measurement Method:** Serial blood sampling and central PK laboratory analysis.

7.2 Secondary & Exploratory Endpoints Safety and tolerability endpoints (including incidence of Adverse Events, injection site reactions, and laboratory abnormalities).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring during the clinical unit stay and outpatient visits, tracking vital signs and injection site reactions. **Serious Adverse Events (SAEs):** Standard reporting timeline (typically 24 hours). **Data Monitoring Committee (DMC):** Internal Phase 1 safety monitoring and review.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Pharmacokinetic (PK) Analysis Set; Safety Set for all AE assessments. **Sample Size Justification:** Powered to establish the pharmacokinetic comparability/bridging between the APFS and AI delivery devices. **Handling of Missing Data:** Standard Phase 1 PK imputation rules for blood concentrations below the lower limit of quantification.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Rigorous onsite source data verification at the dedicated Phase 1 clinical units. **Audit Trail:** Robust eCRF data entry with automated query resolution and formal database locking mechanisms.

11. Study Identifiers & Governance

Primary ID: CTIS2024-511840-22-00 **Registry Number:** NCT06908577
Sponsor/Lead Organization: AstraZeneca **Study Title:** An Accessorised Prefilled Syringe to an Autoinjector Pharmacokinetic Bridging Study of Tozorakimab **Official Regulatory Title:** An Accessorised Prefilled Syringe to an Autoinjector Pharmacokinetic Bridging Study of Tozorakimab

12. Clinical Framework (Study Specific)

Primary Condition: Healthy Participants **Diagnosis Threshold:** Healthy Adults with BMI between 19 and 30 kg/m2 **Medical Methodology:** Device Delivery Evaluation (Autoinjector vs.

13. Study Procedures & Methodology

Management Pathway: Intensive Phase 1 Clinical Unit Protocol
Education Protocol (HCP): Specialized training on the SC
administration procedures for both the APFS and AI devices
Education Protocol (Patient): Informed consent details regarding
the injection site (abdomen, thigh, or upper arm) and strict
contraception compliance **Quality Audit Mechanism:** Continuous
onsite safety evaluations and central lab PK parameter auditing

14. Observation & Follow-Up Schedule

Total Duration: 16 weeks (113 days of follow-up) **Onsite Visit
Cadence:** Extensive initial clinical unit stay (up to 9 days),
with sequential outpatient follow-ups extending to Day 85 and a
final assessment at Day 113 **Remote Monitoring Frequency:** N/A
(Standardized Phase 1 direct clinical follow-up) **Checklist Review
Interval:** Continuous AE, ECG, and injection site status review
throughout the 16-week post-dose period

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				